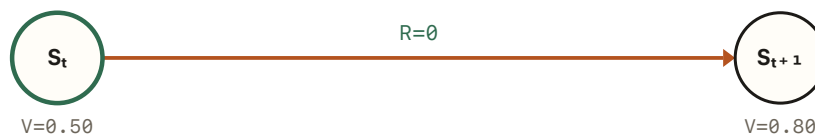


TD, Sarsa & Q-Learning — Problems

Problem 1. TD(0) PREDICTION

A single TD(0) prediction update. Estimate the new value of state S_t after one transition.

Given: $V(S_t) = 0.50$, reward $R_{t+1} = 0$, $\gamma = 0.9$, $V(S_{t+1}) = 0.80$, step size $\alpha = 0.1$.



One step of experience: agent moves $S_t \rightarrow S_{t+1}$ collecting reward 0.

Problem 2. TD ERROR & TARGET

Identify the TD target and the TD error for a single transition.

Given: $R_{t+1} = 1$, $\gamma = 0.95$, $V(S_{t+1}) = 10$, $V(S_t) = 6$.

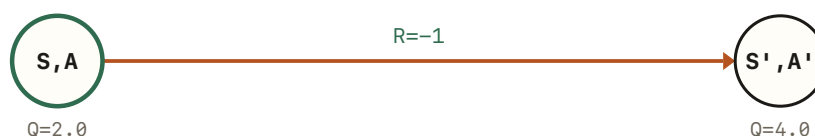


The target blends the observed reward with the bootstrapped estimate of the next state.

Problem 3. SARSA (ON-POLICY)

Perform one Sarsa update for the action-value $Q(S_t, A_t)$.

Given: $Q(S_t, A_t) = 2.0$, $R_{t+1} = -1$, $\gamma = 0.9$, next state-action $Q(S_{t+1}, A_{t+1}) = 4.0$, $\alpha = 0.5$.



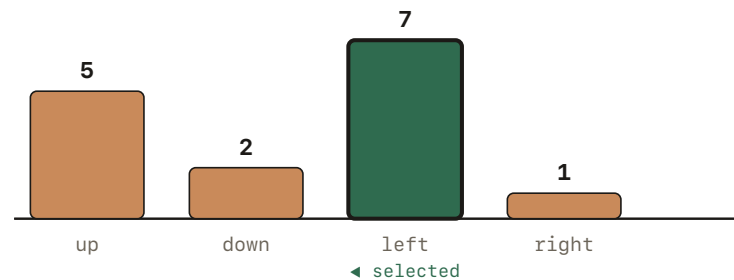
Sarsa uses the tuple (S, A, R, S', A') — it bootstraps off the action actually taken next.

Problem 4. Q-LEARNING (OFF-POLICY)

Perform one Q-learning update. From the next state there are four available actions.

Given: $Q(S,A) = 3.0$, $R = 0$, $\gamma = 1.0$, $\alpha = 0.2$. Next-state values: $Q(S',up)=5$, $Q(S',down)=2$, $Q(S',left)=7$, $Q(S',right)=1$.

$Q(S', \cdot)$ – Q-learning takes the max regardless of policy



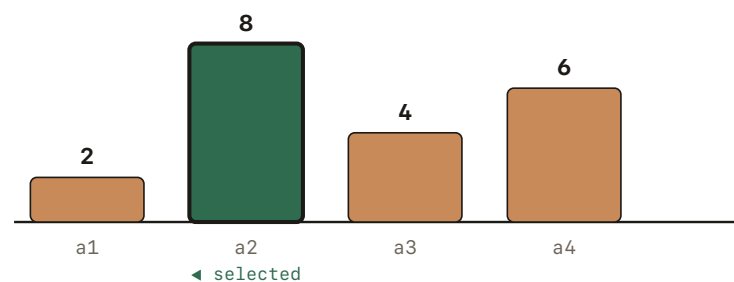
Off-policy: the target uses $\max_a Q(S',a)$ = the greedy action, even if the agent won't take it.

Problem 5. EXPECTED SARSA

Perform one Expected Sarsa update using an ϵ -greedy policy ($\epsilon = 0.1$) over 4 actions.

Given: $Q(S', \cdot)$: $a1=2$, $a2=8$, $a3=4$, $a4=6$. Greedy action = $a2$. $Q(S,A)=5.0$, $R=1$, $\gamma=0.9$, $\alpha=0.1$.

$Q(S', \cdot)$ – Expected Sarsa weights ALL actions by policy prob

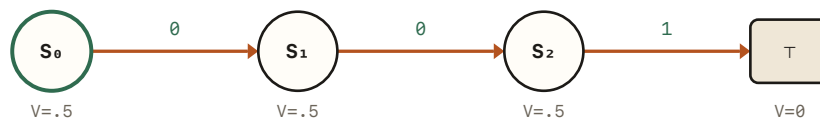


Instead of one sampled A' , Expected Sarsa averages $Q(S',a)$ weighted by $\pi(a|S')$.

Problem 6. MC ERROR = Σ TD ERRORS

Verify that the Monte-Carlo error equals the sum of TD errors over a 3-step episode ($\gamma = 1$, V held fixed).

Given: Episode $S_0 \rightarrow S_1 \rightarrow S_2 \rightarrow \text{terminal}$ with rewards 0, 0, 1. Estimates $V(S_0) = V(S_1) = V(S_2) = 0.5$, $V(\text{terminal}) = 0$.

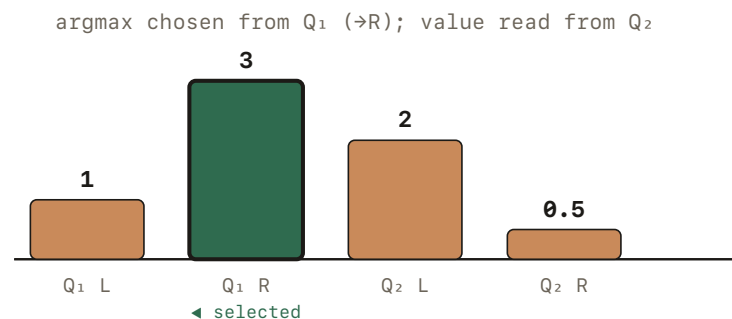


Random-walk style episode: the terminal reward of 1 propagates back through the TD errors.

Problem 7. DOUBLE Q-LEARNING

Perform one Double Q-learning update to Q_1 , using the standard decoupled max.

Given: State S' with 2 actions. $Q_1(S',L)=1$, $Q_1(S',R)=3$. $Q_2(S',L)=2$, $Q_2(S',R)=0.5$. $Q_1(S,A)=2.0$, $R=0$, $\gamma=0.9$, $\alpha=0.5$.

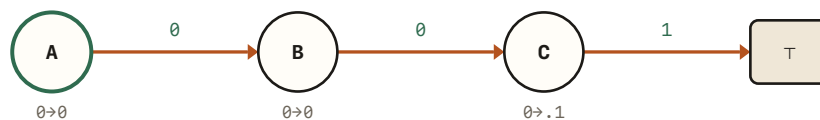


Decoupling: Q_1 selects the action, Q_2 supplies its value — this removes maximization bias.

Problem 8. TD(0) OVER AN EPISODE

Run TD(0) for one full episode over a 3-state chain and report the updated value table.

Given: States $A \rightarrow B \rightarrow C \rightarrow \text{terminal}$, all V initialised to 0. Rewards: $A \rightarrow B = 0$, $B \rightarrow C = 0$, $C \rightarrow T = 1$. $\alpha = 0.1$, $\gamma = 1.0$. Updates applied online (immediately).

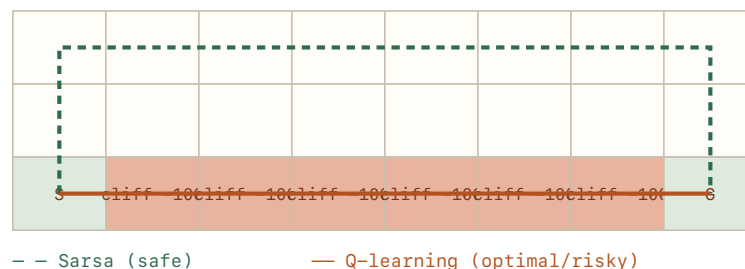


With online TD(0), only the last state changes after one episode — reward has not yet propagated back.

Problem 9. SARSA VS Q-LEARNING (CLIFF)

On the Cliff-Walking edge, compare the TD target of Sarsa vs Q-learning for one step. Which is more pessimistic near the cliff?

Given: From S' , two actions: $Q(S', \text{up}) = -1$ (moves away from cliff), $Q(S', \text{right}) = -100$ (steps toward cliff region). $R = -1$, $\gamma = 1.0$. Sarsa happens to sample the exploratory action $A' = \text{right}$.



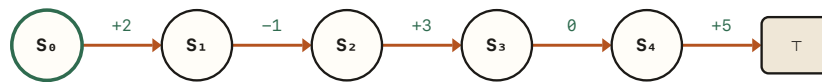
-- Sarsa (safe) — Q-learning (optimal/risky)

Cliff-Walking: Q-learning learns the optimal edge path; Sarsa, feeling exploration's cost, prefers the safe detour.

Problem 10. MULTI-STEP DISCOUNTED RETURN

Compute the full discounted return G_0 from a 5-step reward sequence — the quantity Monte-Carlo TD methods estimate.

Given: Rewards from $t=0$: $R_1=2$, $R_2=-1$, $R_3=3$, $R_4=0$, $R_5=5$. Discount $\gamma = 0.9$.



Return = discounted sum of future rewards; each step is weighted by an extra factor of γ .

Worked Solutions

Solution 1. TD(0) PREDICTION

1. TD update rule.

$$V(S_t) \leftarrow V(S_t) + \alpha [R_{t+1} + \gamma \cdot V(S_{t+1}) - V(S_t)]$$

2. Compute the TD target.

$$R + \gamma \cdot V(S_{t+1}) = 0 + 0.9 \times 0.80 = 0.72$$

3. Compute the TD error δ .

$$\delta = \text{target} - V(S_t) = 0.72 - 0.50 = 0.22$$

4. Apply the step size.

$$V(S_t) \leftarrow 0.50 + 0.1 \times 0.22 = 0.50 + 0.022$$

✓ $V(S_t) = \mathbf{0.522}$ (TD error $\delta = 0.22$)

Solution 2. TD ERROR & TARGET

1. TD target = observed reward + discounted estimate of next state.

$$\text{target} = R + \gamma \cdot V(S_{t+1}) = 1 + 0.95 \times 10 = 1 + 9.5 = 10.5$$

2. TD error = target – current estimate.

$$\delta = 10.5 - 6 = 4.5$$

Why it's an estimate.

The target itself is an estimate for two reasons: it samples the expected reward, AND it uses the estimated $V(S_{t+1})$ rather than the true value.

✓ TD target = **10.5**, TD error $\delta = \mathbf{4.5}$

Solution 3. SARSA (ON-POLICY)**1. Sarsa update rule (uses the actual next action A_{t+1}).**

$$Q(S,A) \leftarrow Q(S,A) + \alpha [R + \gamma \cdot Q(S',A') - Q(S,A)]$$

2. Compute the target.

$$R + \gamma \cdot Q(S',A') = -1 + 0.9 \times 4.0 = -1 + 3.6 = 2.6$$

3. TD error.

$$\delta = 2.6 - 2.0 = 0.6$$

4. Apply update.

$$Q \leftarrow 2.0 + 0.5 \times 0.6 = 2.0 + 0.3$$

$$\checkmark Q(S_t, A_t) = 2.3$$

Solution 4. Q-LEARNING (OFF-POLICY)**1. Q-learning update rule.**

$$Q(S,A) \leftarrow Q(S,A) + \alpha [R + \gamma \cdot \max_a Q(S',a) - Q(S,A)]$$

2. Find the max over next-state actions.

$$\max_a Q(S',a) = \max(5, 2, 7, 1) = 7 \quad (\text{action} = \text{left})$$

3. Compute the target.

$$R + \gamma \cdot 7 = 0 + 1.0 \times 7 = 7$$

4. TD error and update.

$$\delta = 7 - 3.0 = 4.0 \rightarrow Q \leftarrow 3.0 + 0.2 \times 4.0 = 3.0 + 0.8$$

$$\checkmark Q(S,A) = 3.8$$

Solution 5. EXPECTED SARSA**1. ϵ -greedy action probabilities ($\epsilon=0.1$, 4 actions).**

Non-greedy each: $\epsilon/4 = 0.025$. Greedy (a_2): $1 - \epsilon + \epsilon/4 = 0.9 + 0.025 = 0.925$

2. Expected value of next state under π .

$E = 0.025 \cdot 2 + 0.925 \cdot 8 + 0.025 \cdot 4 + 0.025 \cdot 6 = 0.05 + 7.4 + 0.10 + 0.15 = 7.70$

3. Target = $R + \gamma \cdot E$.

$1 + 0.9 \times 7.70 = 1 + 6.93 = 7.93$

4. Update.

$\delta = 7.93 - 5.0 = 2.93 \rightarrow Q \leftarrow 5.0 + 0.1 \times 2.93 = 5.0 + 0.293$

✓ Expected next value = 7.70, $Q(S,A) = \mathbf{5.293}$

Solution 6. MC ERROR = Σ TD ERRORS

1. TD error at each step: $\delta_t = R_{t+1} + \gamma V(S_{t+1}) - V(S_t)$.

$$\delta_0 = 0 + 0.5 - 0.5 = 0$$

$$\delta_1 = 0 + 0.5 - 0.5 = 0$$

$$\delta_2 = 1 + 0 - 0.5 = 0.5$$

2. Sum of TD errors.

$$\Sigma \delta_t = 0 + 0 + 0.5 = 0.5$$

3. Monte-Carlo error at S_0 ($\gamma=1 \Rightarrow$ return $G_0 = 0+0+1 = 1$).

$$G_0 - V(S_0) = 1 - 0.5 = 0.5$$

4. Conclusion.

$$\Sigma \delta_t = 0.5 = G_0 - V(S_0) \checkmark \quad (\text{the identity holds when } V \text{ is not updated mid-episode})$$

✓ Both equal **0.5** — the MC error decomposes exactly into the sum of TD errors.

Solution 7. DOUBLE Q-LEARNING**1. Update rule (updating Q_1).**

$$Q_1(S, A) \leftarrow Q_1(S, A) + \alpha [R + \gamma \cdot Q_2(S', \arg\max_a Q_1(S', a)) - Q_1(S, A)]$$

2. Pick the greedy action using Q_1 .

$$\arg\max_a Q_1(S', a): Q_1(S', L)=1, Q_1(S', R)=3 \rightarrow a^* = R$$

3. Evaluate that action using Q_2 .

$$Q_2(S', a^*) = Q_2(S', R) = 0.5$$

4. Target and update.

$$\begin{aligned} \text{target} &= 0 + 0.9 \times 0.5 = 0.45; & \delta &= 0.45 - 2.0 = -1.55 \\ Q_1 &\leftarrow 2.0 + 0.5 \times (-1.55) = 2.0 - 0.775 \end{aligned}$$

✓ $Q_1(S, A) = 1.225$

Solution 8. TD(0) OVER AN EPISODE**Step A→B (R=0).**

$$V(A) \leftarrow 0 + 0.1[0 + 1 \cdot V(B) - V(A)] = 0 + 0.1[0 + 0 - 0] = 0$$

Step B→C (R=0).

$$V(B) \leftarrow 0 + 0.1[0 + 1 \cdot V(C) - V(B)] = 0 + 0.1[0 + 0 - 0] = 0$$

Step C→T (R=1, V(T)=0).

$$V(C) \leftarrow 0 + 0.1[1 + 1 \cdot 0 - 0] = 0 + 0.1 \times 1 = 0.1$$

Key insight.

Because C was updated AFTER A and B were already read as 0, the reward only reaches C this episode. It propagates one state further per subsequent episode – the hallmark of one-step bootstrapping.

✓ **$V(A) = 0$, $V(B) = 0$, $V(C) = 0.1$**

Solution 9. SARSA VS Q-LEARNING (CLIFF)**1. Q-learning target (uses the max — ignores exploration).**

$$R + \gamma \cdot \max(Q(S', \text{up}), Q(S', \text{right})) = -1 + 1 \cdot \max(-1, -100) = -1 + (-1) = -2$$

2. Sarsa target (uses the actually sampled A' = right).

$$R + \gamma \cdot Q(S', \text{right}) = -1 + 1 \cdot (-100) = -101$$

3. Interpretation.

Sarsa's target is far more pessimistic (-101) because it accounts for the real chance that ϵ -greedy exploration will step off the cliff. Q-learning's target (-2) reflects only the optimal greedy continuation.

4. Consequence.

Sarsa therefore learns to avoid the edge (safer, longer path); Q-learning learns the optimal but risky edge path and takes more penalties online.

✓ Q-learning target = **-2**; Sarsa target = **-101**. Sarsa is more pessimistic $\Rightarrow$ safer path.

Solution 10. MULTI-STEP DISCOUNTED RETURN**1. Return definition.**

$$G_0 = R_1 + \gamma R_2 + \gamma^2 R_3 + \gamma^3 R_4 + \gamma^4 R_5$$

2. Discount factors ($\gamma=0.9$).

$$\gamma^0=1, \gamma^1=0.9, \gamma^2=0.81, \gamma^3=0.729, \gamma^4=0.6561$$

3. Weighted terms.

$$1 \cdot 2 = 2.0$$

$$0.9 \cdot (-1) = -0.9$$

$$0.81 \cdot 3 = 2.43$$

$$0.729 \cdot 0 = 0$$

$$0.6561 \cdot 5 = 3.2805$$

4. Sum.

$$G_0 = 2.0 - 0.9 + 2.43 + 0 + 3.2805 = 6.8105$$

**✓ $G_0 = 6.8105$**